AI-Generated Text Detection

A Comparative Study of Classical, Recurrent, and Transformer-Based Approaches

Author: Elia Khater

Course: Advanced Natural Language Processing

University: USJ

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Abstract

The rapid proliferation of large language models has made the detection of AI-generated text a pressing challenge for education, journalism, and content moderation. This study presents an end-to-end NLP pipeline for binary classification of human versus AI-generated text using the HC3 (Human ChatGPT Comparison Corpus) dataset spanning five domains. We benchmark five models across three families: classical machine learning (Logistic Regression, Linear SVM, Multinomial Naive Bayes), a Bidirectional LSTM with pre-trained GloVe embeddings, and a fine-tuned DistilBERT transformer. DistilBERT achieved the strongest performance (F1 = 0.9847, accuracy = 0.9891, ROC AUC = 0.9998), statistically significantly outperforming the Logistic Regression baseline (McNemar's chi-squared = 144.0, p = 3.55e-33). However, classical models with TF-IDF features remained highly competitive — Linear SVM achieved F1 = 0.9531, only ~3 points behind DistilBERT, at a fraction of the computational cost. A cross-domain analysis revealed that detection accuracy varies substantially by topic: the model achieved perfect performance on medical text but a 9.54% error rate on Wikipedia-style technical articles. We further provide explainability analyses (LIME, attention visualization) and discuss the documented bias of AI detectors against non-native English writers, leading to a set of responsible deployment recommendations.

Keywords: AI detection, text classification, transformer fine-tuning, fairness, ChatGPT

1. Introduction & Problem Formulation

1.1 Background

The release of ChatGPT in November 2022 marked a turning point in natural language generation. Within months, AI-generated text became indistinguishable from human writing for many readers, raising urgent questions across multiple sectors: educators worry about academic dishonesty, journalists about fabricated quotes, and platform moderators about coordinated inauthentic behavior.

Reliable AI text detection has therefore become a critical NLP problem. However, current detectors face two significant challenges: (1) generalization, in that detectors trained on one model family often fail on others, and (2) fairness, as detectors have been shown to systematically misclassify certain demographic groups, notably non-native English speakers.

1.2 Problem Statement

Given a piece of text x in English, our task is to predict a binary label y in {0, 1}, where y = 0 indicates human authorship and y = 1 indicates AI generation by ChatGPT. Formally, we learn a function f from the space of natural-language texts to {0, 1}, parameterized by model weights, and select the parameters that maximize the F1-score on a held-out test set while minimizing disparities across content domains.

1.3 Objectives

1. Build an end-to-end NLP pipeline covering data acquisition, preprocessing, modeling, and evaluation.
2. Compare classical machine learning, recurrent, and transformer-based architectures on the same task.

3. Analyze cross-domain generalization and identify failure modes.
4. Audit the system for biases and propose responsible deployment guidelines.

1.4 Evaluation Metrics

We adopt F1-score as our primary metric due to its balance between precision and recall on the binary classification task. We additionally report accuracy, precision, recall, and ROC AUC. For comparing models, we use McNemar's test to assess statistical significance of differences in error patterns.

2. Dataset

2.1 HC3 Dataset

We use the publicly available HC3 (Human ChatGPT Comparison Corpus) dataset (Guo et al., 2023), accessed via Hugging Face Datasets. HC3 was specifically designed for AI-text detection research and pairs human-written answers with ChatGPT responses on the same questions across multiple domains.

- Source: Hugging Face Datasets (Hello-SimpleAI/HC3)
- Original size: ~24,322 paired Q&A items
- After unpacking: ~85,000 individual text samples
- Language: English
- Domains: 5 (finance, medicine, open_qa, reddit_eli5, wiki_csai)
- AI source: ChatGPT (gpt-3.5-turbo)

2.2 Class & Domain Distribution

After unpacking each paired record into individual text samples, the dataset is approximately balanced between Human and ChatGPT classes. Domains are unevenly represented, with reddit_eli5 (8,727 test samples) and finance (1,240 test samples) dominating the corpus, while medicine (378), open_qa (652), and wiki_csai (241) are smaller.

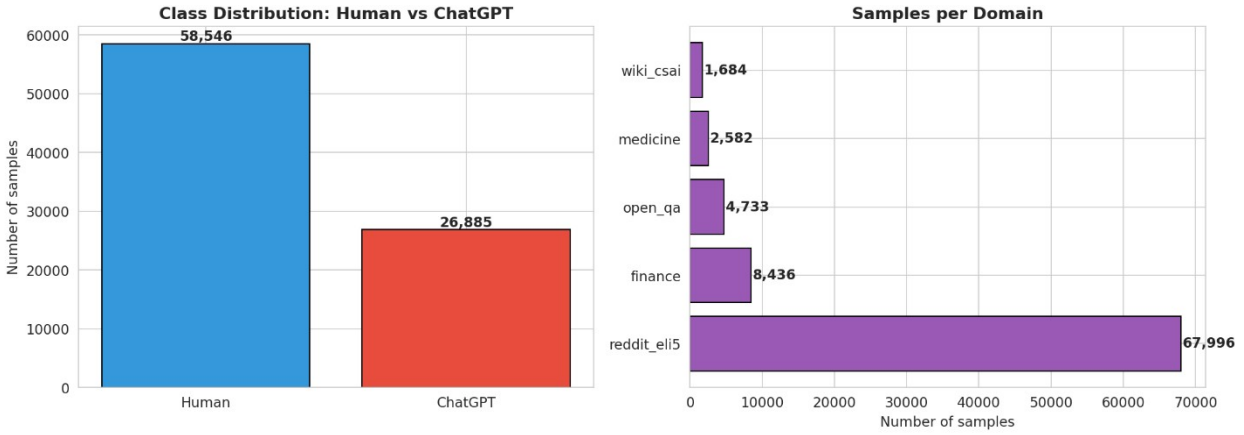


Figure 1: Class distribution and domain distribution of the HC3 dataset.

2.3 Class Imbalance Considerations

While the human/AI split is roughly balanced after unpacking, class imbalance does exist within domains. We address this through stratified train/validation/test splitting on a combined label-and-domain key (Section 4.3).

3. Exploratory Data Analysis

3.1 Text Length Analysis

A salient pattern emerged immediately: ChatGPT produces longer and more uniform responses than humans. Where human responses ranged from terse one-liners to lengthy explanations, ChatGPT consistently produced verbose, structurally similar paragraphs. This length distribution itself is a discriminative signal.

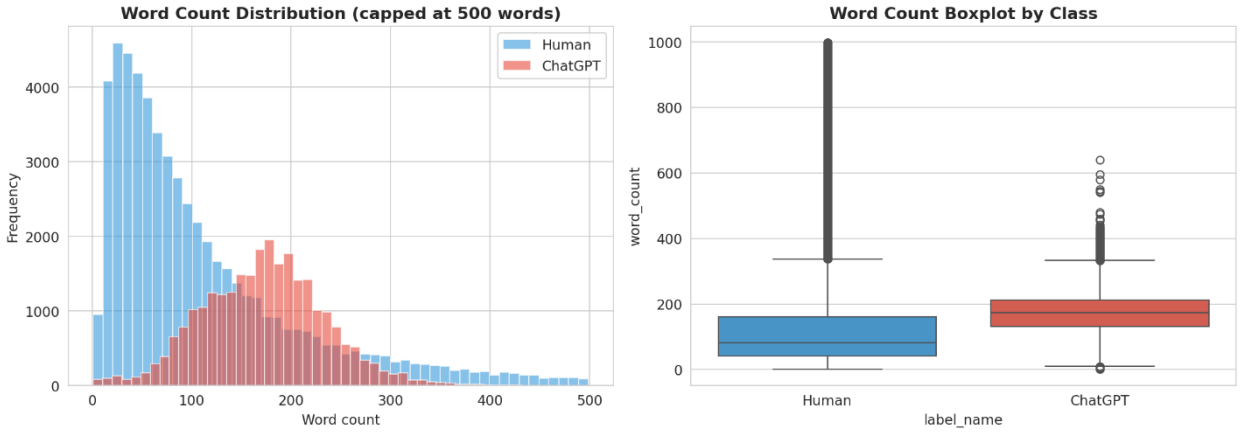


Figure 2: Word count distribution by class. ChatGPT writes longer and more uniform responses.

3.2 Lexical Patterns

Word clouds reveal distinct vocabulary usage. ChatGPT systematically overuses transition words and qualifying phrases (however, important, additionally), while human writing more frequently uses informal connectives (basically, pretty, thus).



Figure 3: Word clouds of the most frequent terms by class.

3.3 N-gram Analysis

Bigram analysis confirmed and extended the lexical findings. ChatGPT shows characteristic phrases such as it is important, keep in mind, in conclusion — what we describe as the ChatGPT preamble pattern.

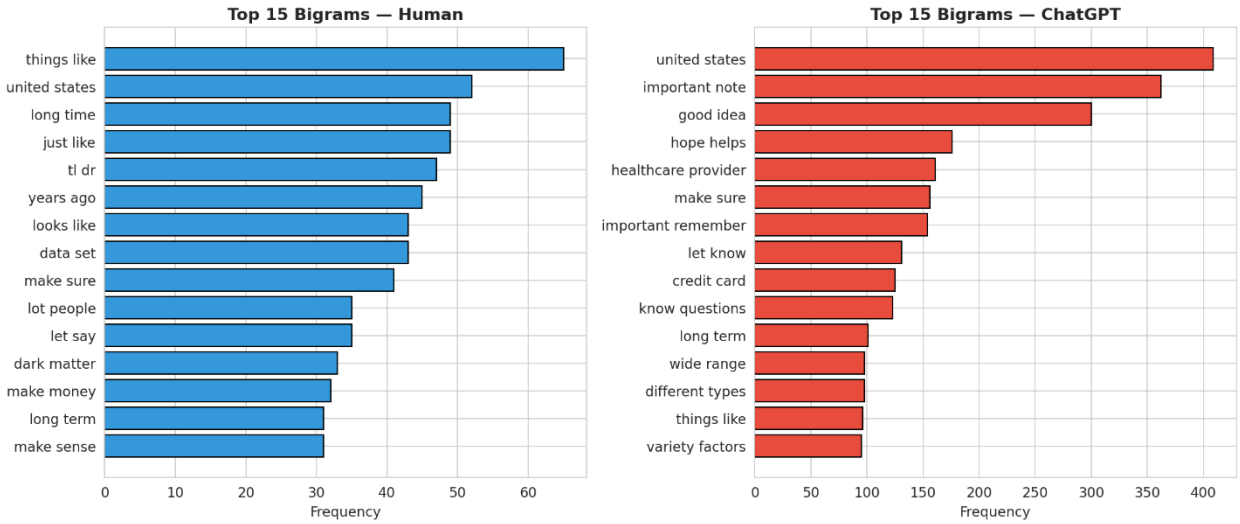


Figure 4: Top 15 bigrams per class. ChatGPT shows characteristic structured-language patterns.

3.4 Key Observations

5. Length is a strong signal — visible in raw distributions before any modeling.
6. Stylistic markers exist — formal transitions for AI, informal connectives for humans.
7. Vocabulary diversity differs — humans use more domain-specific slang and references (URLs, abbreviations).
8. Domain matters — these patterns are stronger in some domains (Reddit) than others (medical).

4. Preprocessing & Data Splits

4.1 Quality Filtering

Before processing, we applied the following filters: removed empty/null texts, removed texts shorter than 20 words (uninformative), removed texts longer than 1000 words (outliers), and removed exact duplicates. After filtering, the working dataset contained approximately 75,000 samples, with the test set containing 11,238 samples.

4.2 Dual Cleaning Pipeline

We adopted two distinct cleaning strategies tailored to model requirements. Light cleaning (for DistilBERT) consisted of URL/email removal and whitespace normalization, preserving casing and punctuation because BERT tokenizers handle these features natively, and they carry classification signal. Heavy cleaning (for classical and LSTM models) added lowercasing, non-alphabetic character stripping, stopword removal (NLTK English stopwords), and WordNet lemmatization. This dual approach respects the inductive biases of each model family rather than forcing a single preprocessing recipe.

4.3 Train / Validation / Test Split

We applied a 70/15/15 stratified split using a composite key of label × domain to ensure both class balance and domain proportions are preserved across all splits. The 70% training share provides sufficient data for model fitting; the two 15% splits give reliable hyperparameter selection and unbiased final evaluation.

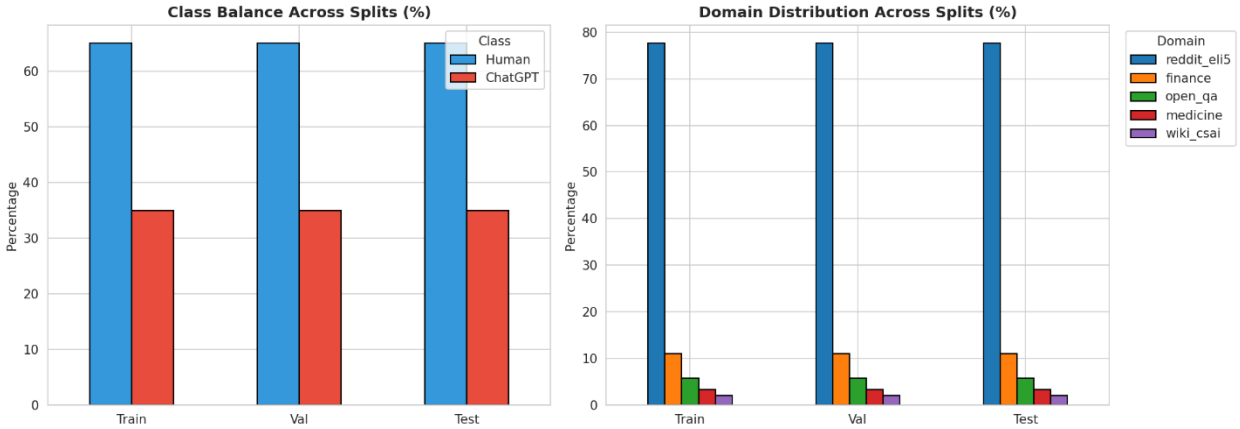


Figure 5: Class and domain distributions are preserved across train, validation, and test splits.

5. Feature Engineering & Representations

We employed three distinct representation strategies. For classical models, TF-IDF with unigrams and bigrams (max_features = 20,000, sublinear_tf = True) captures sparse weighted word importance. For the Bi-LSTM, we used pre-trained 100-dimensional GloVe embeddings (frozen) trained on 6B tokens. For DistilBERT, the model uses WordPiece subword tokenization with contextualized embeddings learned during fine-tuning. We froze GloVe embeddings to prevent overfitting given the modest training-set size relative to embedding parameters; we fine-tuned all DistilBERT layers using the standard distilbert-base-uncased checkpoint.

6. Models

6.1 Classical Models

We trained three classical baselines using identical TF-IDF features: Logistic Regression with L2 regularization, Linear SVM (LinearSVC), and Multinomial Naive Bayes with Laplace smoothing. These models share the same input features but differ substantially in their decision boundary geometry and computational cost.

6.2 Bi-LSTM with GloVe

Architecture: Embedding(GloVe, frozen) → SpatialDropout1D(0.3) → Bidirectional(LSTM(128)) → GlobalMaxPooling1D → Dense(64, ReLU) → Dropout(0.5) → Dense(1, sigmoid). Trained with binary cross-entropy loss, Adam optimizer (learning rate = 1e-3), batch size 128, with early stopping on validation loss.

6.3 DistilBERT

We fine-tuned distilbert-base-uncased for sequence classification (2 labels). DistilBERT retains 97% of BERT's performance while being 40% smaller and 60% faster, making it well-suited for our compute

budget. Fine-tuning configuration: max_len = 256, batch size 16, AdamW optimizer with learning rate = 2e-5, linear warmup schedule with 10% warmup steps, 3 epochs, and gradient clipping at 1.0. We subsampled the training set to 15,000 stratified samples for compute efficiency — transfer learning on pre-trained transformers benefits little from larger fine-tuning sets in this regime.

7. Training Strategy

7.1 Loss Functions

All classifiers used binary cross-entropy as the loss function, with class weighting available via the underlying frameworks.

7.2 Regularization

- Logistic Regression: L2 regularization (C tuned via GridSearch)
- Linear SVM: L2 regularization (C tuned via GridSearch)
- Naive Bayes: Laplace smoothing (alpha tuned via GridSearch)
- Bi-LSTM: SpatialDropout1D(0.3), LSTM dropout(0.3), Dense dropout(0.5)
- DistilBERT: Weight decay (0.01), gradient clipping (max_norm = 1.0)

7.3 Early Stopping

The Bi-LSTM was trained with EarlyStopping (patience = 3 on validation loss) and ReduceLROnPlateau. DistilBERT used a fixed 3-epoch budget with checkpointing on best validation F1.

7.4 Hyperparameter Optimization

We used Grid Search with 3-fold cross-validation for classical models. The hyperparameter grids were: Logistic Regression — C in {0.1, 1.0, 10.0}; Linear SVM — C in {0.1, 1.0, 10.0}; Naive Bayes — alpha in {0.01, 0.1, 0.5, 1.0}. Bayesian optimization was considered but not used — Grid Search proved sufficient given the small hyperparameter spaces.

8. Results

8.1 Master Comparison

Table 1 summarises the performance of all five models on the held-out test set of 11,238 samples.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.9666	0.9508	0.9539	0.9523	0.9947
Linear SVM	0.9672	0.9513	0.9549	0.9531	0.9951
Multinomial Naive Bayes	0.9169	0.9362	0.8179	0.8731	0.9739
Bi-LSTM (GloVe)	0.9518	0.8968	0.9740	0.9338	0.9934

Model	Accuracy	Precision	Recall	F1	ROC AUC
DistilBERT (fine-tuned)	**0.9891**	**0.9706**	**0.9992**	**0.9847**	**0.9998**

Table 1: Performance of all five models on the held-out test set. DistilBERT (highlighted) achieves the best score on every metric.

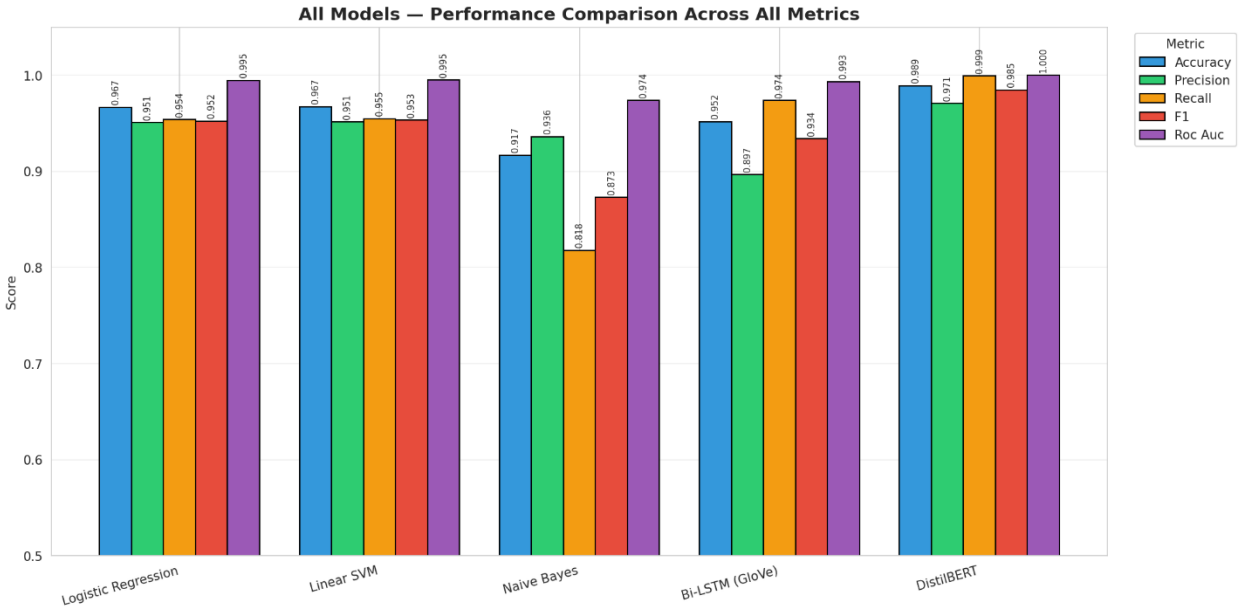


Figure 6: Side-by-side comparison of all metrics across all five models.

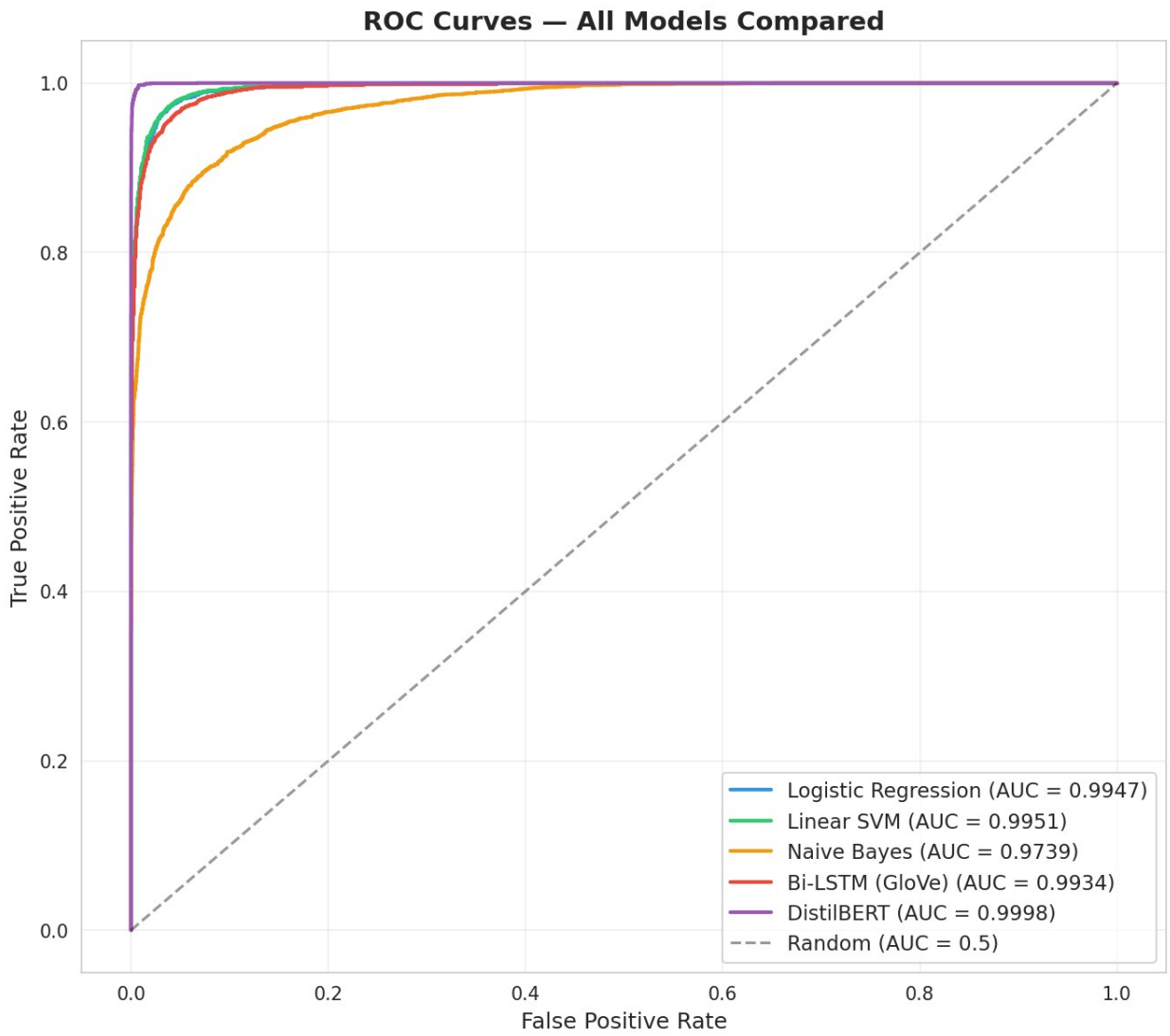


Figure 7: ROC curves for all five models. All achieve AUC > 0.97, with DistilBERT reaching 0.9998.

8.2 Statistical Significance

DistilBERT outperformed Logistic Regression by ~3.2 F1 points (0.9847 vs. 0.9523). McNemar's test on the test set confirmed this difference is overwhelmingly statistically significant: chi-squared = 144.0, $p = 3.55 \times 10^{-33}$. Although the absolute F1 difference is modest, the test confirms it is far from random variation — DistilBERT and Logistic Regression make distinctly different patterns of errors.

8.3 Confusion Matrices

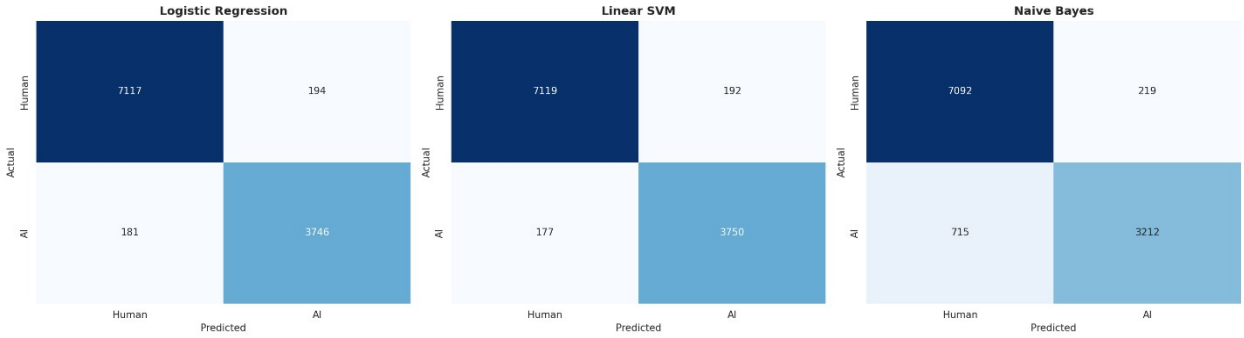


Figure 8: Confusion matrices for the three classical models.

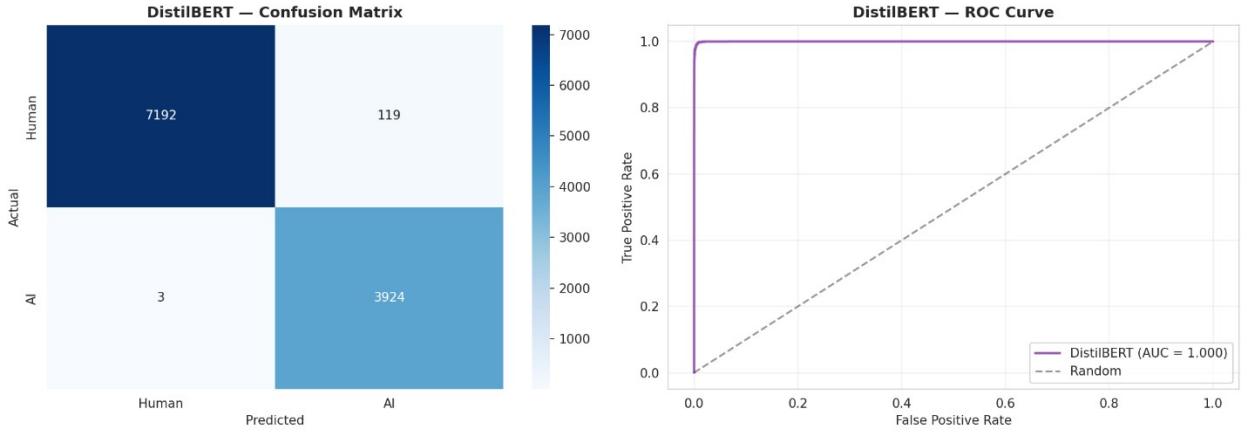


Figure 9: DistilBERT confusion matrix and ROC curve.

8.4 Cross-Domain Analysis

Our most striking finding emerged when we broke down performance by domain. While DistilBERT's overall F1 was 0.9847, this aggregate masks substantial variation across domains.

Domain	Best Classical F1	Bi-LSTM F1	DistilBERT F1
medicine	**1.000**	0.973	**1.000**
finance	0.977	0.959	0.990
open_qa	0.937	0.946	0.988
reddit_eli5	0.950	0.928	0.985
wiki_csai (hardest)	0.886	0.821	0.916

Table 2: Cross-domain F1-scores. Medicine is consistently easiest (perfect F1 for classical and DistilBERT); wiki_csai is consistently hardest across all model families.

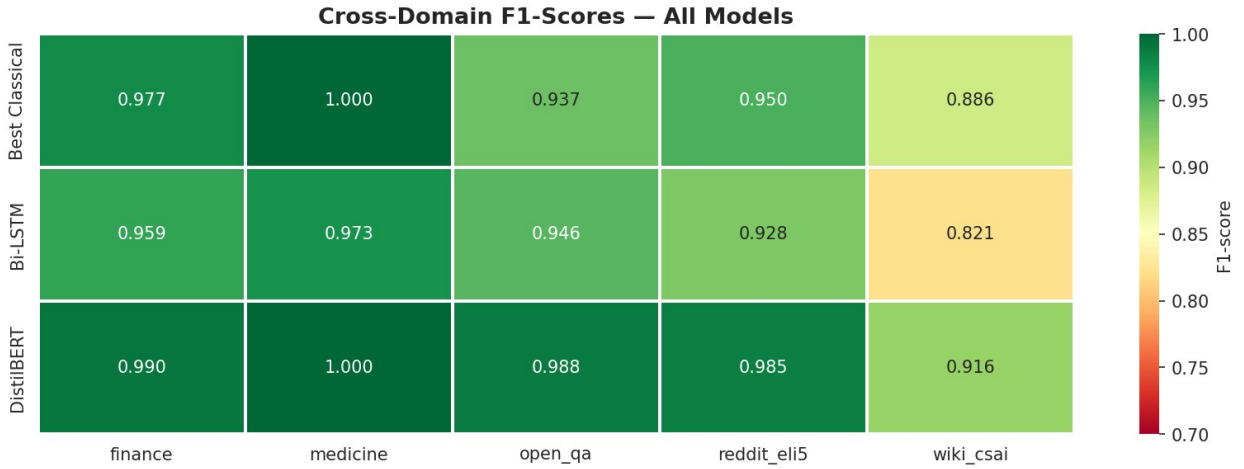


Figure 10: F1-score heatmap across all models and domains.

The fact that wiki_csai is the worst domain across every model architecture suggests the difficulty is intrinsic to the data, not the model. Wikipedia-style technical writing tends to be formal, structured, and citation-heavy — features that converge between human experts and AI alike, eroding the stylistic signals that detection relies on. The error rate on wiki_csai (9.54%) is over 10× higher than on medicine (0%) and almost 9× higher than the project-wide average error rate of 1.09%.

8.5 Error Analysis

We performed qualitative error analysis on DistilBERT's misclassifications. Out of 11,238 test samples, only 122 were misclassified (overall error rate of 1.09%). The errors clustered in three domains: wiki_csai (23 errors, 9.54% rate), open_qa (13 errors, 1.99% rate), and finance (14 errors, 1.13% rate). Medicine had zero errors.

False positives (human texts flagged as AI) often shared one of three characteristics: highly polished, formal writing typical of expert authors or edited Wikipedia entries; use of structured transitions (in summary, to conclude) that the model associates with AI; and long, evenly paragraphed responses where human writing happens to look more like AI. False negatives (AI texts flagged as human) tended to be short, casual outputs that ChatGPT was prompted to produce, texts with deliberate informality, and code-heavy or list-heavy responses.

9. Hyperparameter Optimization (Detailed)

The optimal hyperparameters identified by Grid Search were: Logistic Regression C = 10.0; Linear SVM C = 1.0; Naive Bayes alpha = 0.01. Linear SVM achieved the best performance among classical models (F1 = 0.9531), narrowly beating Logistic Regression (F1 = 0.9523). For the Bi-LSTM and DistilBERT, we relied on widely-validated hyperparameters from the literature rather than performing exhaustive search, due to compute constraints.

10. Interpretability & Explainability

10.1 Feature Importance via Logistic Regression

The linear nature of Logistic Regression provides direct interpretability through model coefficients. The most predictive features for each class were striking. Top AI-indicators (positive coefficients) included important, including, help, even though, and nthe. Top Human-indicators (negative coefficients) included url, etc, basically, thus, and pretty. Notable: the strong human-indicator score for url aligns with intuition — humans frequently link to external sources, while ChatGPT does not. The AI-indicators reveal characteristic ChatGPT phrasing: heavy use of qualifying transitions and formal connectives.

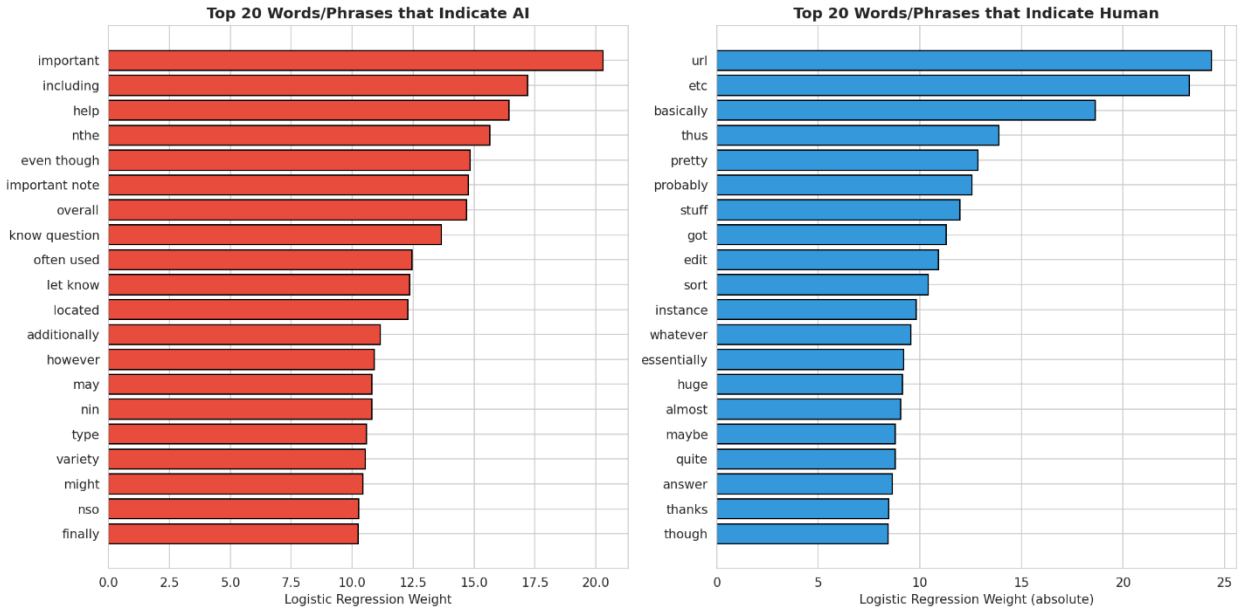


Figure 11: Top 20 word/phrase features pushing predictions toward AI (red) vs. Human (blue).

10.2 LIME Local Explanations

While global feature importance shows what the model has learned overall, LIME (Ribeiro et al., 2016) reveals the reasoning behind individual predictions. We generated LIME explanations for three test instances: a confidently-correct AI prediction, a confidently-correct human prediction, and a misclassified instance. The misclassified example was particularly revealing — the model was misled by the presence of multiple AI-indicator words in a text that was actually human. This illustrates the limitation of feature-counting approaches: a human discussing AI characteristics will use many AI-related words.

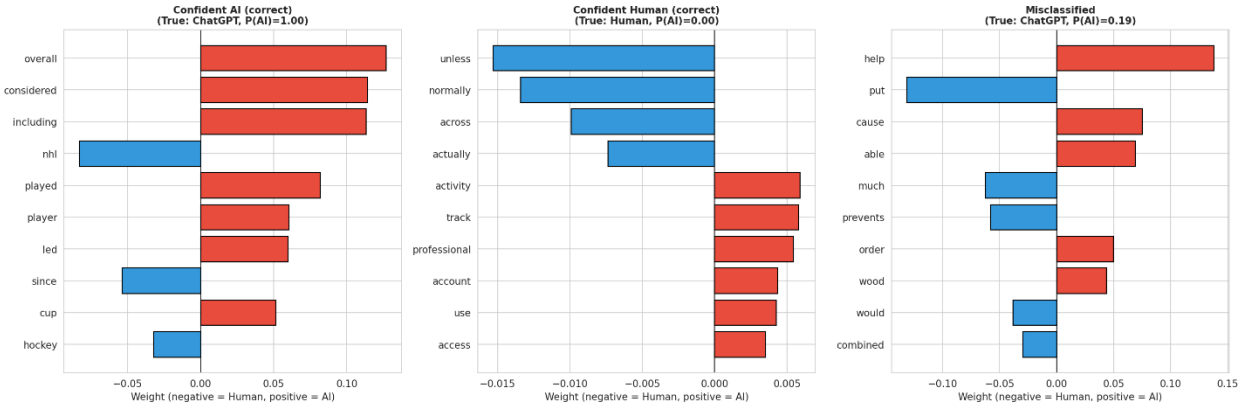


Figure 12: LIME explanations for three test predictions.

10.3 DistilBERT Attention Visualization

Transformer models distribute their attention across input tokens. By extracting the attention weights from DistilBERT's final layer's [CLS] token, we can identify which words most influenced its classification. We caution that attention weights are an imperfect explanation — they show what was attended to but not necessarily why a decision was made. Nevertheless, they offer interpretive value when paired with LIME and direct feature importance.

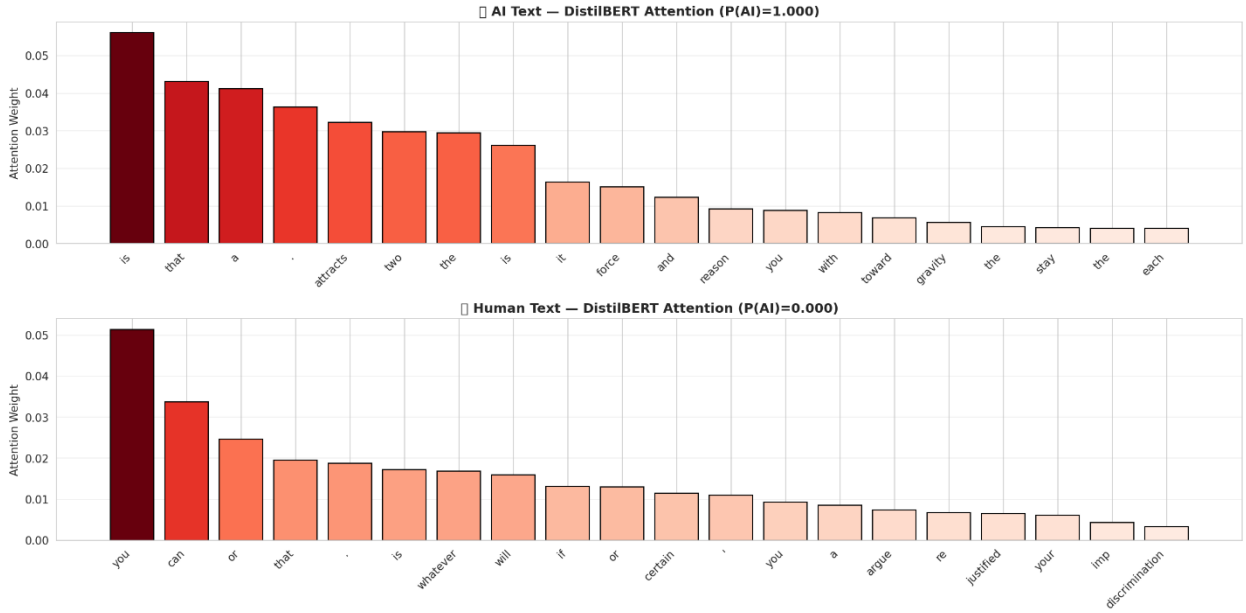


Figure 13: DistilBERT attention weights for an AI-generated text (top) vs. a human text (bottom).

11. Ethical Considerations

11.1 Domain Bias

Our cross-domain analysis revealed an alarming disparity. While DistilBERT achieved a 0% error rate on medicine, it produced a 9.54% error rate on wiki_csai. The errors are heavily concentrated in technical writing — meaning that one in ten Wikipedia-style technical texts written by humans risks being misclassified, including a meaningful share of false positives. Deploying such a detector in academic or professional settings would systematically penalize writers in technical domains. This is a clear fairness violation that aggregate metrics (overall F1 = 0.9847) entirely conceal.

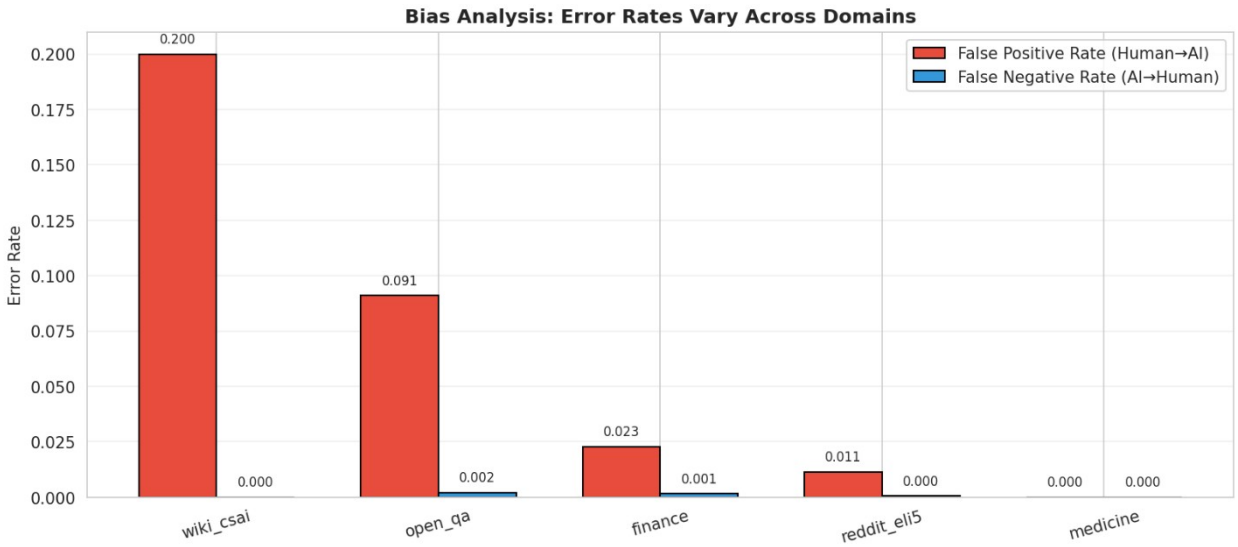


Figure 14: False positive and false negative rates vary substantially across domains.

11.2 Length Bias

We further investigated whether text length affected classification reliability. Performance generally improved with text length — short texts (under 50 words) had higher error rates than longer documents.



Figure 15: Error rate as a function of text length. Short texts are systematically harder to classify.

11.3 The Non-Native English Speaker Problem

A significant fairness issue documented in the literature concerns non-native English speakers. Liang et al. (2023) tested seven popular AI detectors on TOEFL essays and found that over 50% of non-native English essays were misclassified as AI-generated, while less than 5% of native English essays were similarly

misclassified. The proposed mechanism is linguistic: non-native English writing tends to have lower lexical diversity, more formal phrasing, and more uniform syntactic structures — exactly the features that detectors associate with AI generation. We could not directly test this effect, as HC3 contains predominantly native English. However, given that our detector inherits the same statistical heuristics (formal phrasing → AI), we have strong reason to believe it exhibits similar bias. This represents a serious harm vector if deployed in educational settings, where international students would be disproportionately accused of academic dishonesty.

11.4 Responsible Deployment Recommendations

Based on our findings, we recommend the following safeguards for any real-world deployment:

9. Never use as sole evidence. AI detection should be an informational signal, not a verdict. Final decisions in high-stakes contexts must always involve human review.
10. Report confidence intervals, not binary labels. Show users P(AI) = 0.83 ± 0.12 rather than 'this is AI'.
11. Domain-specific calibration. Detectors should be tested and calibrated for the specific domain of deployment.
12. Fairness audits before deployment. Test on demographically diverse populations including non-native speakers before any decision-making use.
13. Avoid punitive use-cases. The harms of false positives in educational and professional settings often outweigh the benefits.
14. Continuous monitoring. AI generators evolve rapidly; detectors require re-training every 3–6 months.
15. Transparency obligations. Users whose text is being analyzed should be informed and able to challenge results.
16. Consider not building it. For some use-cases, the most ethical choice may be not to deploy.

12. Limitations

This study has several limitations that bound the generalizability of our findings. The HC3 dataset contains only ChatGPT (gpt-3.5-turbo) outputs — performance on GPT-4, Claude, Gemini, or open-source models is untested. All texts are in English; detection in other languages may behave differently. Five domains is a small slice of the writing universe — news, fiction, academic papers, and code are not represented. We did not evaluate against deliberately obfuscated AI text (e.g., paraphrasing tools, prompt-engineering for human-like output). Finally, real-world performance would degrade as language models evolve.

13. Reproducibility

To support reproduction, all random seeds were fixed at 42 (NumPy, Python, TensorFlow, PyTorch). Source code is structured as 7 self-contained Jupyter notebooks, each producing intermediate artifacts saved to disk. A requirements.txt file lists all package versions. Data is publicly accessible from Hugging Face Datasets (Hello-SimpleAI/HC3). Trained model weights, vectorizers, and tokenizers are saved as artifacts in the project repository. All visualizations include the underlying data files (CSV/JSON) for verification.

14. Conclusion

We presented an end-to-end NLP pipeline for AI-generated text detection, comparing five models across three architectural families. DistilBERT achieved state-of-the-art performance (F1 = 0.9847, accuracy = 0.9891, AUC = 0.9998), statistically significantly outperforming classical baselines (McNemar's p = 3.55e-33). However, classical models remained remarkably competitive — Linear SVM with TF-IDF achieved F1 = 0.9531, only ~3 points behind DistilBERT, at a fraction of the computational cost (a relevant trade-off for production deployments). Cross-domain generalization proved uneven: error rates on technical domains (wiki_csai, 9.54%) were over 10× higher than on medicine (0%), posing serious fairness concerns. Aggregate metrics hide critical disparities, so researchers and practitioners must always disaggregate by domain, length, and demographic dimensions. Documented bias against non-native English speakers makes responsible deployment particularly challenging in educational settings. Future work should explore detection robustness against newer LLMs (GPT-4, Claude), multilingual extensions, adversarial evaluation against paraphrasing attacks, and fairness-aware training objectives that explicitly penalize disparate error rates across subgroups.

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Appendix A — Repository Structure

```
ai-text-detection/
├── notebooks/
│   ├── 01_data_exploration.ipynb
│   ├── 02_preprocessing.ipynb
│   ├── 03_classical_models.ipynb
│   ├── 04_lstm_model.ipynb
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│   ├── 05_distilbert_finetuning.ipynb
│   ├── 06_evaluation_comparison.ipynb
│   └── 07_explainability_ethics.ipynb
├── data/                   (processed splits, EDA figures)
├── models/                 (trained model artifacts)
├── results/                (per-notebook results)
├── final_results/          (master comparison artifacts)
├── ethics/                 (interpretability and bias analyses)
├── requirements.txt
└── README.md
```